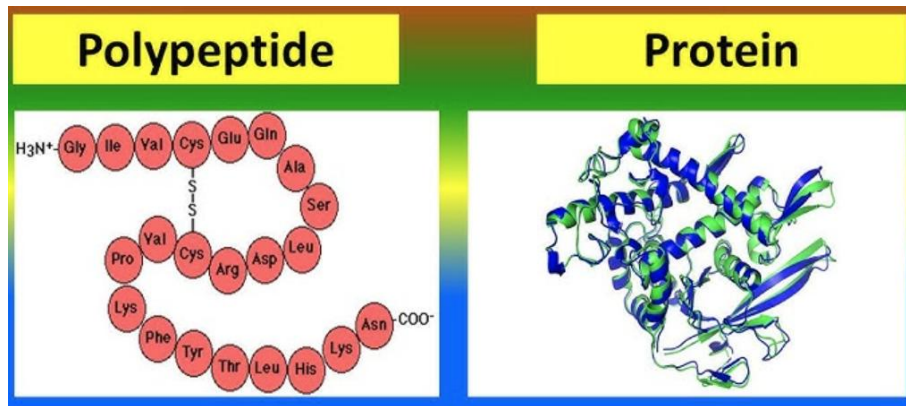


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Ch 4 Enzymes

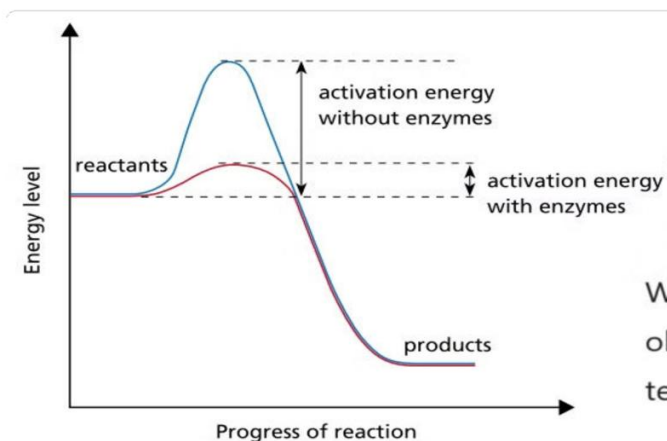
Enzymes 酶 are proteins that catalyze 催化 reactions



Polypeptide 多肽	►	Protein 蛋白質
Formed by amino acid 胺基酸 condensation	Folding (Hydrogen bonds between amino acids)	Formed by one or more polypeptides
Linear 線性 / chain		3D conformation determined by amino acid sequence 胺基酸序列
Inactive		Specific 獨特 function

Enzymes catalyze reaction by lower activation energy 活化能

Enzymes 酶 serve as **biological catalysts 催化劑** in our body, providing a reaction pathway with **lower activation energy** so that life-sustaining chemical reactions could take place at body temperatures. ←



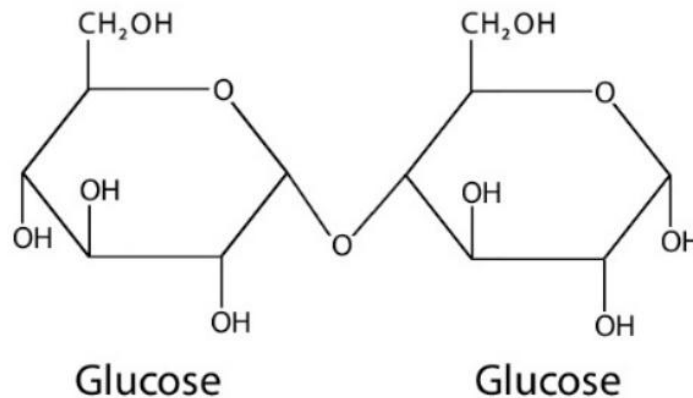
Without enzymes, energy could not be obtained from glucose at body temperatures (~37°C).

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Enzyme working mechanism

example: glucose 葡萄糖 and maltose 麥芽糖

Maltose



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	Reaction
Maltose to glucose	Condensation (building-up) / Hydrolysis (breaking down)

Hydrolysis reaction:

Maltase 麥芽糖酶

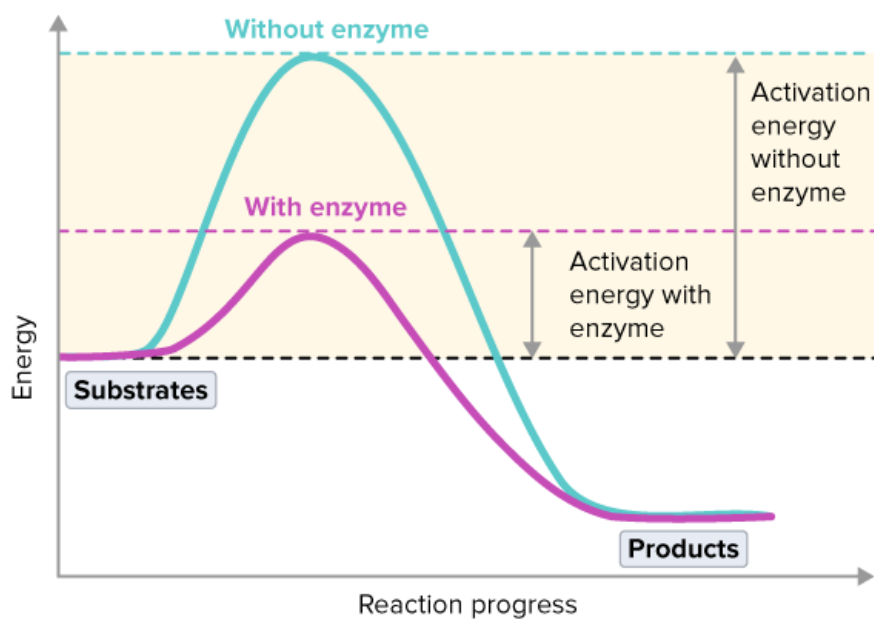
Maltose 麥芽糖 + 1 water

----->

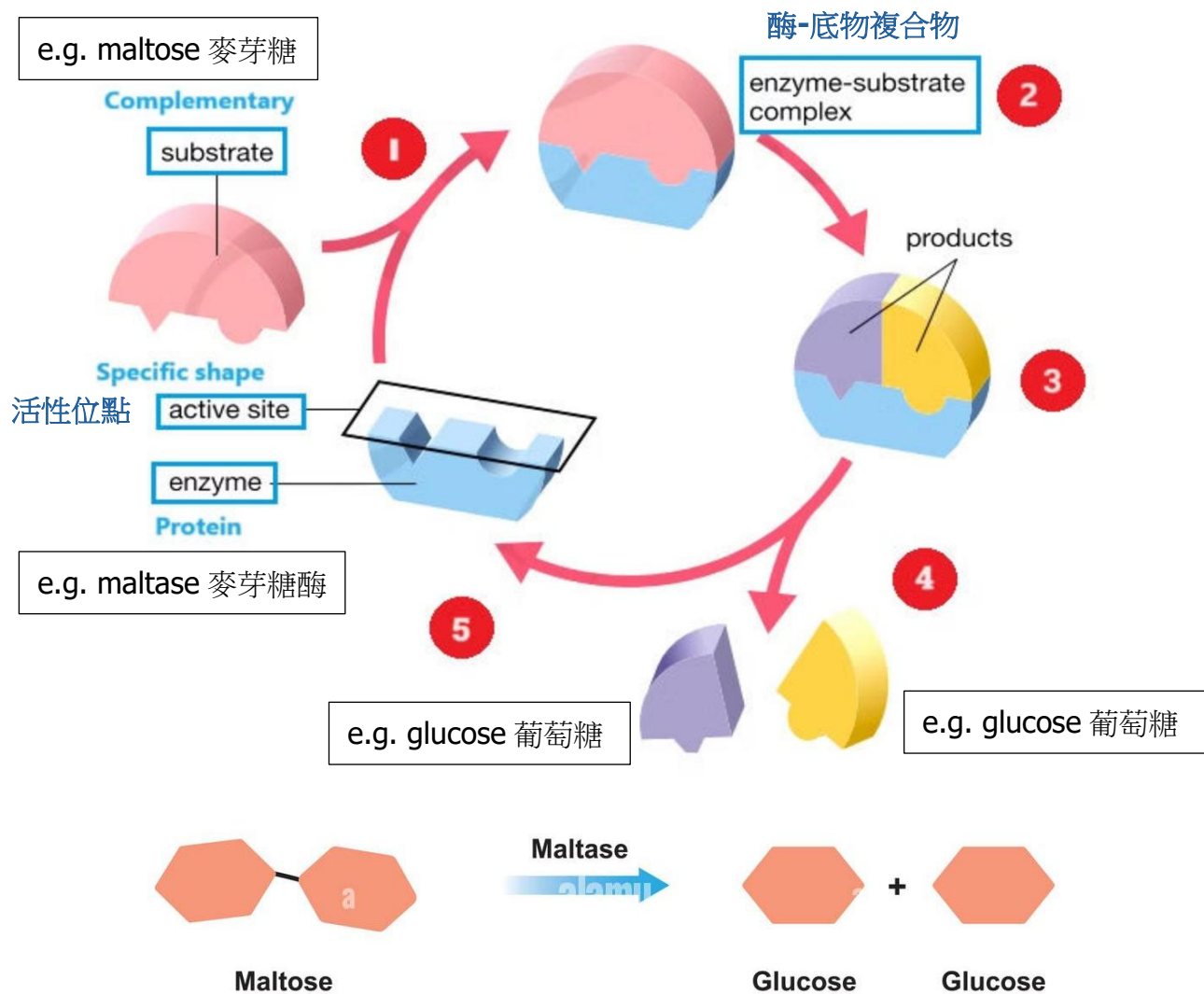
glucose 葡萄糖 + glucose 葡萄糖

Substrates 底物 (入)

Products 產物 (出)



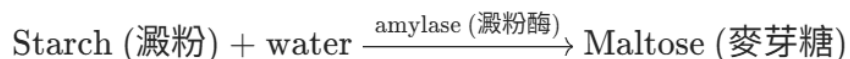
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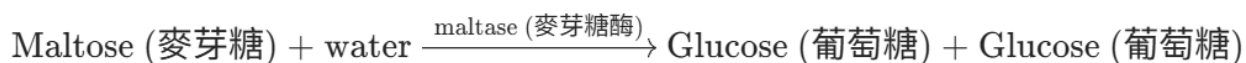
1. <u>Substrate</u> bind to specific 獨特 active site 活性位點 of <u>enzyme</u> by random collision 隨機碰撞	<u>Maltose 麥芽糖</u> bind to specific 獨特 active site of <u>maltase 麥芽糖酶</u> by random collision 隨機碰撞
2. Formation of enzyme-substrate complex 酶底物複合物 lower activation energy	Formation of enzyme-substrate complex lower activation energy
3. Reaction completes, release <u>products</u> (enzyme remains unchanged)	Reaction completes, release <u>2 glucoses</u> <u>葡萄糖</u> (maltase unchanged)

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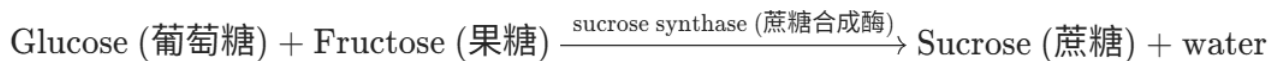
Carbohydrases 碳水化合物酶



Reaction nature	Condensation 縮合 / Hydrolysis 水解 Building up / breaking down
Substrate	
Product	
Enzyme	



Reaction nature	Condensation 縮合 / Hydrolysis 水解
Substrate	
Product	
Enzyme	

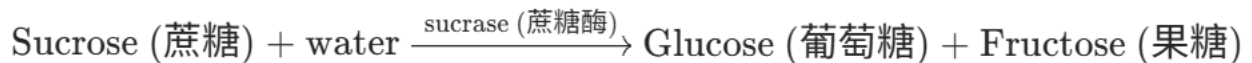


Reaction nature	Condensation 縮合 / Hydrolysis 水解
Substrate	
Product	
Enzyme	

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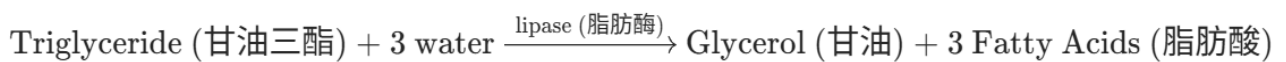


Reaction nature	Condensation 縮合 / Hydrolysis 水解
Substrate	
Product	
Enzyme	



Reaction nature	Condensation 縮合 / Hydrolysis 水解
Substrate	
Product	
Enzyme	

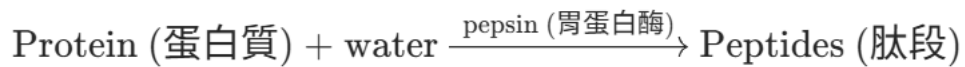
Lipase 脂肪酶



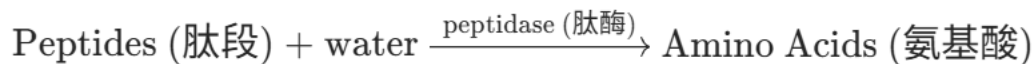
Reaction nature	Condensation 縮合 / Hydrolysis 水解
Substrate	
Product	pH change?
Enzyme	

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Proteases 蛋白酶



Reaction nature	Condensation 縮合 / Hydrolysis 水解
Substrate	
Product	
Enzyme	



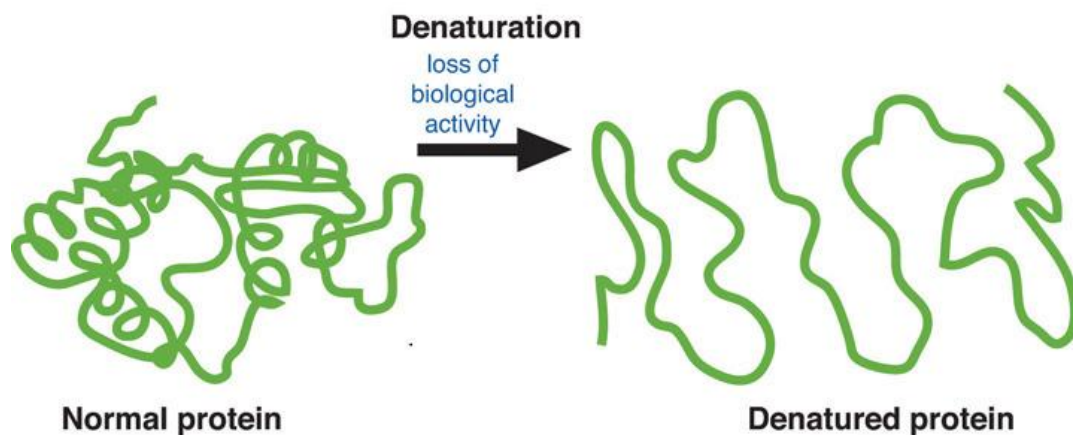
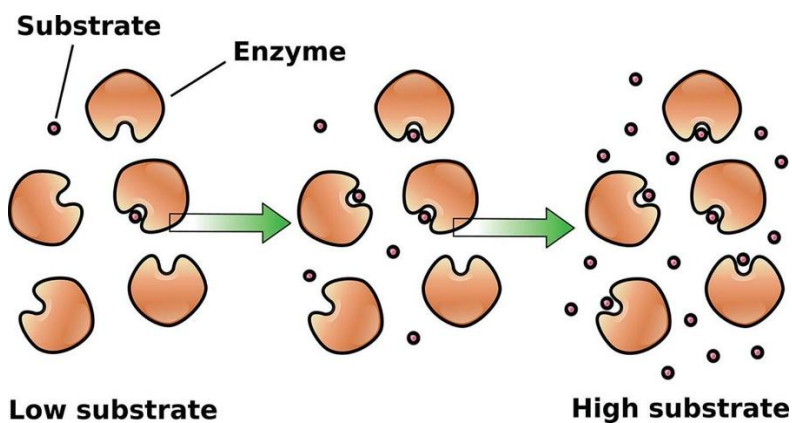
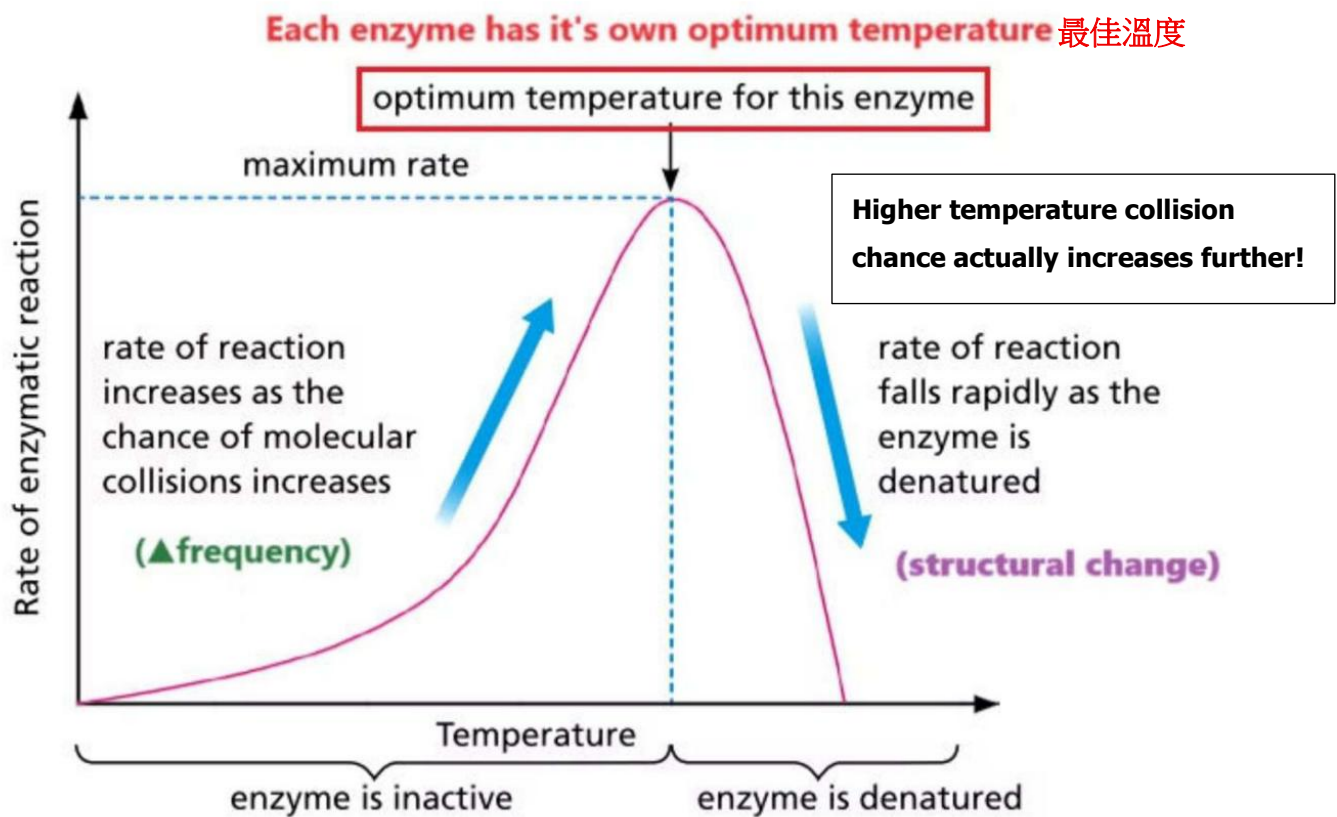
Reaction nature	Condensation 縮合 / Hydrolysis 水解
Substrate	
Product	
Enzyme	



Enzymes are:	Carbohydrates / Lipids / Proteins
Thus, could be broken down by	Carbohydrases / Lipases / Proteases
Active site still intact?	Yes / No
Can form enzyme substrate complex?	Yes / No
Loss of function reversible 可逆? 	Yes / No

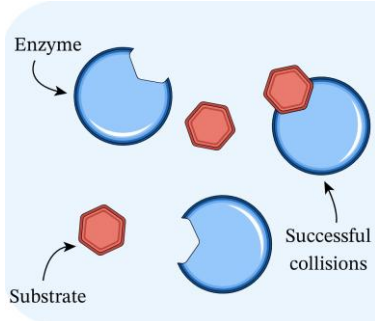
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Effect of temperature on enzymatic reaction rates



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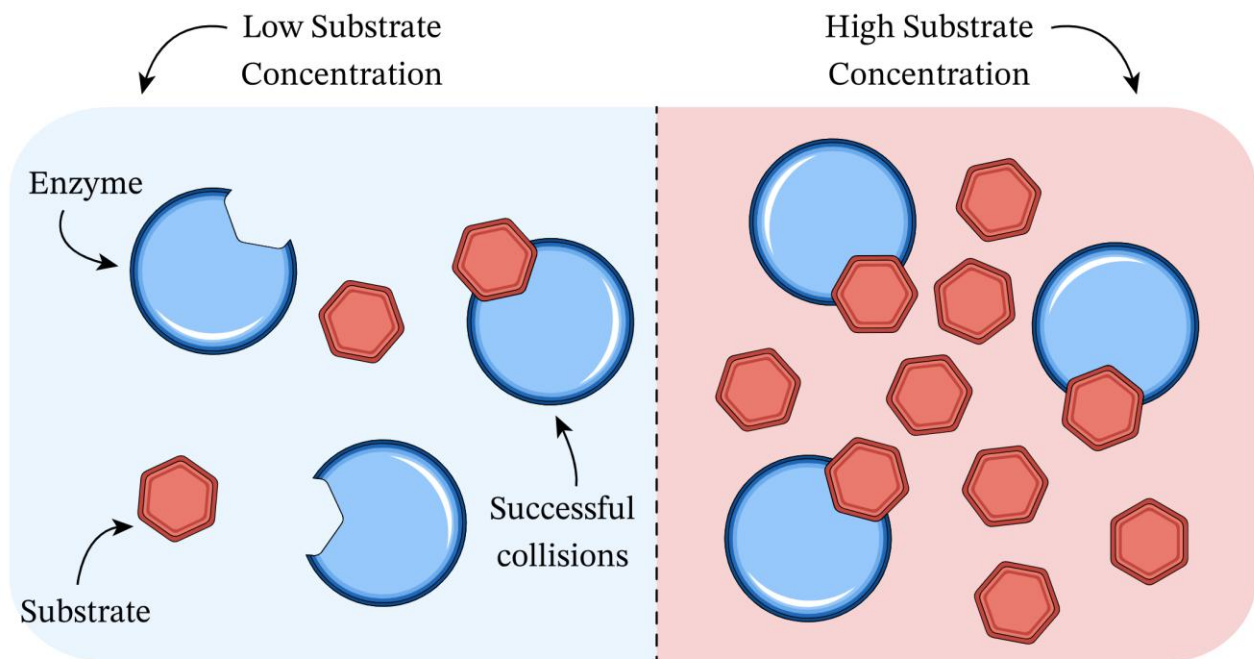
Low temperatures: Inactive 不活躍	
Specific active site 3D conformation 活性位點 3D 構象	Affected / Not affected
Enzyme denatured 變性?	Yes / No
Kinetic energy of substrates and enzymes 動能	High / Low
Collision chance 碰撞機率	High / Low
Chance of forming enzyme-substrate complex	High / Low
Enzymatic reaction rate	High / Low

Increasing temperature to optimum temperature 最佳溫度	
	
Specific active site 3D conformation 活性位點 3D 構象	Affected / Not affected
Enzyme denatured 變性?	Yes / No
Kinetic energy of substrates and enzymes 動能	Higher / Lower
Collision chance 碰撞機率	Higher / Lower
Chance of forming enzyme-substrate complex	Higher / Lower
Enzymatic reaction rate	Higher / Lower

Increasing temperature beyond optimum temperature: denature 變性	
	Is the loss of specific 3D conformation of active sites reversible? Yes / No Can lower temperature back reverse denaturation? Yes / No
Specific active site 3D conformation 活性位點 3D 構象	Affected / Not affected
Enzyme denatured 變性?	Yes / No
Kinetic energy of substrates and enzymes 動能	Higher / Lower
Collision chance 碰撞機率	Higher / Lower
Can form enzyme-substrate complex?	Yes / No
Enzymatic reaction rate	Higher / Lower

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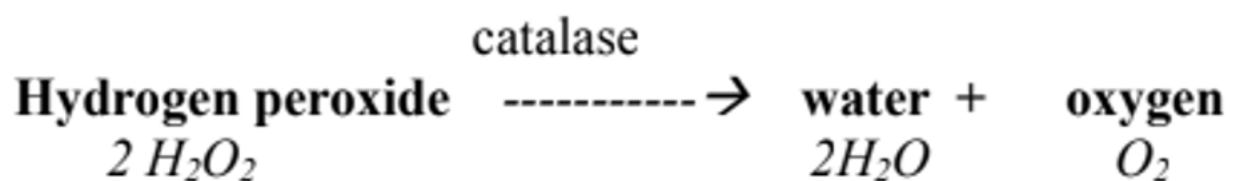
Effect of substrate and enzyme concentrations



Increasing substrate concentration	
Specific active site 3D conformation 活性位點 3D 構象	Affected / Not affected
Enzyme denatured 變性?	Yes / No
Number of active sites	Higher / Lower / no change
Kinetic energy of substrates and enzymes 動能	Higher / Lower / no change
Collision chance 碰撞機率	Higher / Lower / no change
Chance of forming enzyme-substrate complex	Higher / Lower / no change
Enzymatic reaction rate	Higher / Lower / no change

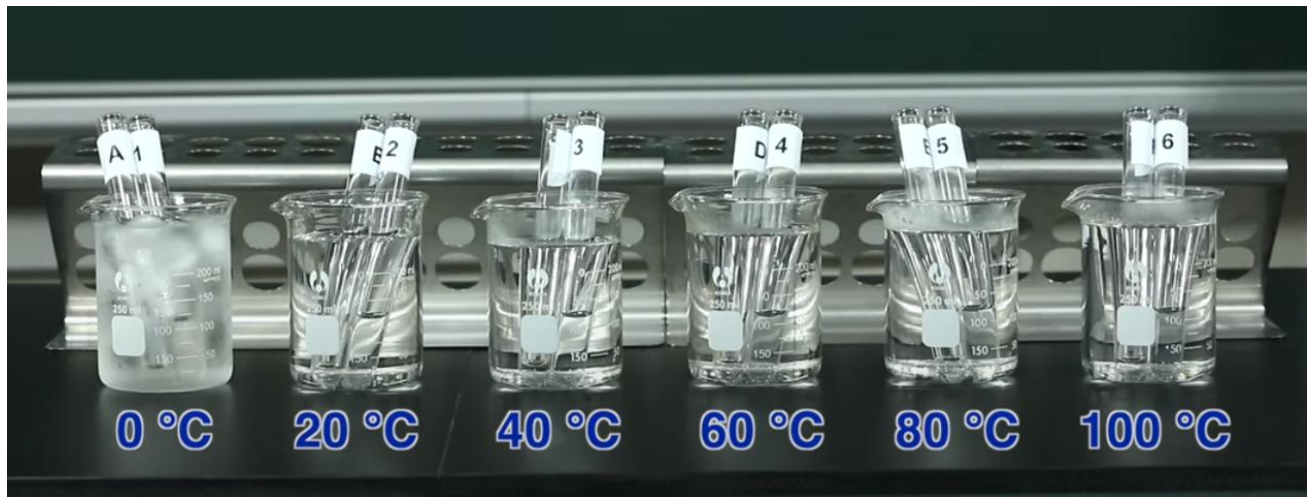
Increasing enzyme concentration	
Specific active site 3D conformation 活性位點 3D 構象	Affected / Not affected
Enzyme denatured 變性?	Yes / No
Number of active sites	Higher / Lower / no change
Kinetic energy of substrates and enzymes 動能	Higher / Lower / no change
Collision chance 碰撞機率	Higher / Lower / no change
Chance of forming enzyme-substrate complex	Higher / Lower / no change
Enzymatic reaction rate	Higher / Lower / no change

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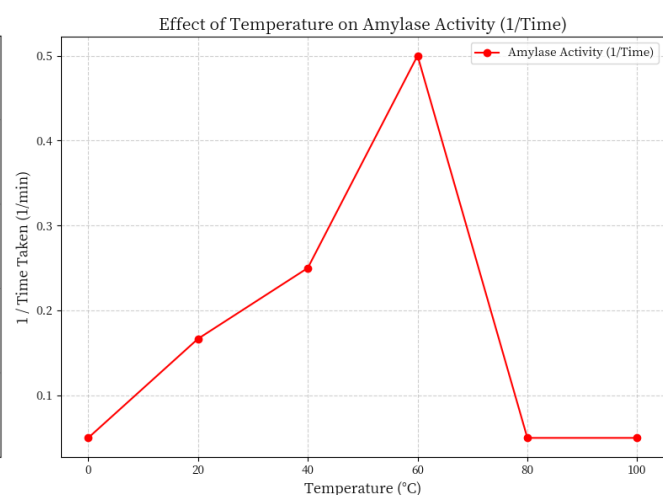
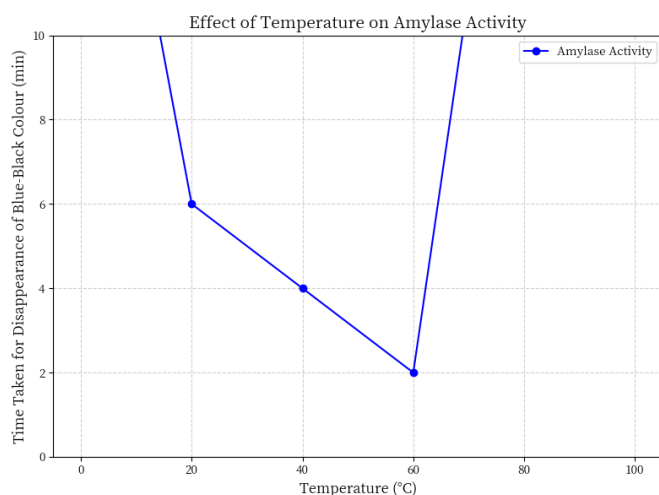
	Increase substrate conc. More hydrogen peroxide 過氧化氫	Increase enzyme conc. More catalase 過氧化氫酶 (e.g. cut potato into small pieces)
Number of active sites	More / Less / no change	More / Less / no change
Kinetic energy of substrates and enzymes 動能	Higher / Lower / no change	Higher / Lower / no change
Collision chance 碰撞機率	Higher / Lower / no change	Higher / Lower / no change
Chance of forming enzyme-substrate complex	Higher / Lower / no change	Higher / Lower / no change
Enzymatic reaction rate	Higher / Lower / no change	Higher / Lower / no change
Oxygen release	More / Less / no change	More / Less / no change

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Starch (澱粉) + water $\xrightarrow{\text{amylase (澱粉酶)}}$ Maltose (麥芽糖)

Temperature (°C)	Time taken for the disappearance of blue-black colour in iodine test (min)	Amount of Starch	Amylase activity
0	The blue-black colour does not disappear. (infinite 無限 time)	No change	Inactive / active / denatured
20	6	Reduced	Inactive / active / denatured
40	4	Reduced	Inactive / active / denatured
60	2	Reduced	Inactive / active / denatured
80	The blue-black colour does not disappear. (infinite time)	No change	Inactive / active / denatured
100	The blue-black colour does not disappear. (infinite time)	No change	Inactive / active / denatured



Optimum temperature is AROUND 60°C!

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Concept check

Multiple Choice Questions (MCQs)

1. What is the primary role of enzymes in biological reactions?
 - A. Increase the energy needed for a reaction
 - B. Provide energy for a reaction
 - C. Lower the activation energy of a reaction
 - D. Convert products into substrates

2. Which of the following is true about enzymes?
 - A. They are consumed in a reaction
 - B. They remain unchanged after a reaction
 - C. They increase the activation energy of a reaction
 - D. They only work at high temperatures

3. What is the substrate for the enzyme maltase?
 - A. Glucose
 - B. Maltose
 - C. Water
 - D. Starch

4. What are the products of the enzymatic breakdown of maltose?
 - A. Glucose and water
 - B. Maltase and water
 - C. Two glucose molecules
 - D. Starch and water

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5. Which statement best describes the active site of an enzyme?
 - A. It is a random region on the enzyme
 - B. It has a specific shape complementary to the substrate
 - C. It is destroyed after each reaction
 - D. It binds to any molecule

6. What happens to the enzyme after releasing the products?
 - A. It is denatured
 - B. It remains unchanged
 - C. It is consumed in the reaction
 - D. It becomes inactive permanently

7. What happens to an enzyme if the temperature exceeds its optimum?
 - A. It becomes more active
 - B. Its active site changes shape, leading to denaturation
 - C. It binds to more substrates
 - D. It stops functioning temporarily but recovers

8. Which of the following factors does NOT affect enzyme activity?
 - A. Temperature
 - B. Substrate concentration
 - C. Enzyme concentration
 - D. Light intensity

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9. How does increasing substrate concentration affect enzymatic activity?
- A. It always increases the rate of reaction
 - B. It increases the rate of reaction until all active sites are occupied
 - C. It decreases the rate of reaction
 - D. It has no effect on the rate of reaction
10. What happens to enzyme activity at low temperatures?
- A. It increases due to high collision frequency
 - B. It decreases due to low kinetic energy of molecules
 - C. It remains unchanged
 - D. The enzyme denatures
11. What is the effect of boiling on enzymes?
- A. Enzymes become more efficient
 - B. Enzymes denature and lose their function
 - C. Enzymes bind more strongly to substrates
 - D. Enzymes become inactive temporarily
12. Which of the following is true about enzyme-substrate complexes?
- A. They form randomly without specific binding
 - B. They lower the activation energy of the reaction
 - C. They increase the activation energy of the reaction
 - D. They are consumed during the reaction

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13. What is the function of catalase in the reaction with hydrogen peroxide?
- A. Breaks down hydrogen peroxide into oxygen and glucose
 - B. Breaks down hydrogen peroxide into water and oxygen
 - C. Combines hydrogen peroxide with water
 - D. Converts hydrogen peroxide into starch
14. What happens when enzyme concentration is increased, assuming substrate concentration is constant?
- A. Reaction rate decreases
 - B. Reaction rate increases until all substrates are used
 - C. Reaction rate remains constant
 - D. Enzyme denaturation occurs
15. What is the optimum temperature for most enzymes in the human body?
- A. 25°C
 - B. 37°C
 - C. 50°C
 - D. 60°C
16. Which of the following best describes enzyme denaturation?
- A. Irreversible loss of the enzyme's specific 3D structure
 - B. Temporary inactivation of the enzyme
 - C. Increased enzyme activity
 - D. A reversible process

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17. What happens to the enzyme-substrate complex when the enzyme is denatured?

- A. It forms faster
- B. It cannot form
- C. It remains unaffected
- D. It forms but does not release products

18. Which of the following factors increases the chance of forming enzyme-substrate complexes?

- A. Low enzyme concentration
- B. Low substrate concentration
- C. Higher collision frequency
- D. Denaturation

19. What is the role of water in the enzymatic breakdown of maltose?

- A. It is a product of the reaction
- B. It is a substrate that participates in the reaction
- C. It denatures the enzyme
- D. It has no role in the reaction

20. Which of the following is a characteristic of enzymes?

- A. They are effective at all temperatures
- B. They are specific to their substrates
- C. They require light for activation
- D. They are consumed during reactions

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True/False Questions

Direction: put a "✓" if the sentence is correct and put a "X" on the phrase or words that are wrong.

1. Enzymes lower the activation energy of a reaction.
2. Substrates bind to the active site of an enzyme randomly.
3. Enzymes are permanently changed after a reaction.
4. Increasing temperature beyond the optimum causes enzyme denaturation.
5. Boiling an enzyme makes it more efficient.
6. Enzymes are specific to their substrates.
7. Increasing substrate concentration always increases the reaction rate.
8. Denaturation of an enzyme is reversible.
9. Enzymes are consumed in biological reactions.
10. Catalase breaks down hydrogen peroxide into water and oxygen.

Fill in the Blanks

Choose from the following:

- | | |
|---|--|
| ☐ enzyme-substrate complex 酶-底物複合物 | ☐ activation energy 活化能 |
| ☐ denaturation 變性 | ☐ substrate 底物 |
| ☐ optimum temperature 最佳溫度 | |
| ☐ active site 活性位點 | |

1. Enzymes lower the _____ of a reaction, making it proceed faster.
2. The _____ binds to the enzyme at the active site.
3. The _____ is the region of the enzyme where the substrate binds.
4. The _____ forms when the substrate binds to the enzyme.
5. Enzymes work most efficiently at their _____.
6. Boiling can cause _____, rendering the enzyme inactive.

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Answer Key

MCQs

1. C
2. B
3. B
4. C
5. B
6. B
7. B
8. D
9. B
10. B
11. B
12. B
13. B
14. B
15. B
16. A
17. B
18. C
19. B
20. B

True/False

1. ✓
2. ✗ (Substrates bind specifically, not randomly.)
3. ✗ (Enzymes remain unchanged after a reaction.)
4. ✓
5. ✗ (Boiling denatures enzymes.)
6. ✓
7. ✗ (Only up to a certain point; then it levels off.)
8. ✗ (Denaturation is often irreversible.)
9. ✗ (Enzymes are not consumed; they remain unchanged.)
10. ✓

Fill in the Blanks

1. Activation energy 活化能
2. Substrate 底物
3. Active site 活性位點
4. Enzyme-substrate complex 酶-底物複合物
5. Optimum temperature 最佳溫度
6. Denaturation 變性